

Preliminary Project Proposal

Reducing Manual Triage Inconsistencies Through Explainable AI-Assisted Triage

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Problem Statement

Triage nurses at low-resource Caribbean health centers sort patients using subjective judgment, producing inconsistent prioritization and elevated mistriage rates (Da'Costa et al., 2025; Tyler et al., 2024). Research shows this method misclassifies up to 1.2% of critical patients, while AI-assisted approaches are nearly 27% more accurate than the average nurse (Tyler et al., 2024). Existing AI triage models were built without clinician input and have not been validated on Caribbean patient data, leaving a critical implementation gap in this context (Araouchi & Adda, 2024; Nsubuga et al., 2025). Thus, I propose a 12-week pilot to build and assess whether a clinician-informed, explainable AI triage prototype reduces triage inconsistency compared to standard nurse-led assessment in a simulated Caribbean ED environment.

Literature Review

Araouchi & Adda (2024), in their research article stated that there was a need for improvement in emergency triaging due to the large number of patients, and complexity of each medical case, overwhelming the nurses who have to triage manually. To come to a solution, the researchers built and tested several AI models designed to predict patient triage levels with precision and accuracy using the Korean Triage and Acuity Scale (KTAS). From this method, the results showed that AI can meaningfully assist in triaging; each model attained a reliable score, however, the Stacking Model proved to be the overall best performer in accuracy, precision, recall and F1-score. Although the method brought such good results, limitations such as a small dataset from only one country took away from those results due to lack of generalization. Furthermore, a gap that these researchers acknowledged was that a larger and more diverse data set is needed in addition to the need for the AI's decisions to be explainable to medical staff.

Da'Costa et al. (2025) found that ED overcrowding, inconsistencies in triaging and the inability of traditional systems to adapt to stressful situations are all creating serious risks to patient safety and staff wellbeing. The authors conducted a narrative review of various peer reviewed studies published between 2014 and 2024 that noted data collection, algorithmic models, real time analytics and natural language processing to assess the benefits, challenges and future directions of AI in triaging. AI triaging systems demonstrated the ability to reduce patient wait times by up to 30%, reduce treatment time by 20%, improve consistency of triage decisions and better manage resources during regular operations and high stress events. The authors flagged a gap in the racial and gender bias in existing AI models as a critical unresolved

problem, alongside the absence of legal frameworks for accountability when AI causes harm and insufficient clinical training.

Tyler et al. (2024) noted that the current nurse led triage process is inconsistent, subjective and under severe pressure from overcrowding, staff shortages and rising patient complexity. To further analyze this issue, the authors conducted a scoping review of 29 peer reviewed US journal studies between 2013 and 2023 from reliable online databases that measured the outcomes of using AI in the triaging process. Across all the studies, AI and ML models consistently outperformed the conventional triaging systems, reduced mistriage rates, improved identification of critically ill patients and, in only one case, surpassed average nurse accuracy by 26.9%. Nearly all studies relied on historical patient data from individual hospitals, making it impossible to confirm how these AI models would actually perform in a live clinical environment across different types of hospitals. The authors also identified gaps including the need for further testing across various hospitals and countries, ethical guidelines around data and patient privacy, and greater clinician involvement in the development and adoption of AI triage systems.

Tahernejad et al (2024) in their review spoke about the use of manual triage and paper-based data collection in hospitals by first responders. AI assisted triage outdates, and even surpasses these processes by being able to track a patient's condition over time, reducing human error in high pressure environments and allows for increased retention of detailed patient information. In conducting this review, the authors sources studies from various medical article databases which focused on applying AI to triage in disaster and emergency settings, rather than

routine hospital conditions. Assessing the outcome, they found that AI assisted triage systems allowed first responders to process approximately three times more patients than traditional methods, while AI diagnostic tools achieved up to 94.57% accuracy in identifying respiratory illnesses during the COVID-19 pandemic. Although the outcomes were impressive, nearly all studies were conducted in a simulated disaster environment, rather than real ones. As a gap, the authors mentioned a lack of AI triage systems due to trust issues, insufficient training, and data privacy concerns.

Nsubuga et al. (2025) found that most low and middle income countries utilize the Kampala Trauma Score (KTS) to triage patients. This method of scoring has been debated against due to its limited predictive accuracy, with patients being scored incorrectly and critical cases being missed entirely. This study took place in rural Uganda and used over 4,000 real patient records to train four machine learning models. While only using basic information from the patient records, all four machine learning models outperformed KTS by a significant margin and were far more reliable at correctly distinguishing patients who needed urgent care from those who could wait. Since the study was conducted in a single hospital in Uganda, the models cannot be generalized across many low resource settings with different numbers of patients, or injuries.

Targeted Gaps

Araouchi & Adda (2024) mentioned that AI's decision making in triage needs to be explainable to medical staff so that clinicians can understand and trust its recommendations. What this means is that most AI triage models work by plainly taking in patient data, evaluating it, and handing out a score in return without explaining its process. The clinician is left contemplating the why factor for numerous cases. The AI's plain score and plain justification is not enough for the triage nurse to feel firm about the decision that is being made.

Similarly, Tyler et al. (2024) stated that clinicians need to be far more involved in the design and adoption of technology and AI tools. Since most technological systems are built by software engineers with minimal experience in other fields, this makes for impressive software on paper, but is frustrating to use by the intended audience. The gap here lies with the people being closest to the solution, the clinicians, being consulted minimally, or even none at all. These two gaps complement each other and can be addressed simultaneously.

By consulting with clinicians, and even other professionals in the medical field, such as hospital administrators, medical researchers, and other health officials, a proper, robust solution to manual triage can be made through the ethical use of AI.

Proposed Solution

Over the next 12 weeks, this project will build and test an application designed to help triage nurses in Caribbean emergency departments make faster, more consistent decisions about which patients need care first. The app will take in basic patient information, such as vital signs and the reason for their visit, and produce a recommended priority level. Importantly, alongside every recommendation, the app will explain in plain language exactly why that score was given, so the nurse always understands the reasoning behind it and feels confident acting on it. To make sure the app actually works for the people who will use it, at least two structured conversations will be held with emergency healthcare professionals during the design process. Their feedback will directly shape how the app looks, what it shows, and how it communicates its recommendations. This is the heart of the project, building something with clinicians, not just for them.

By the end of the 12 weeks, the project will produce a working version of the app tested against existing patient data, a visual mock-up of the interface, and full documentation of the design process and clinical input received.

References

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